Evaluation Results +Quantization Error Analysis

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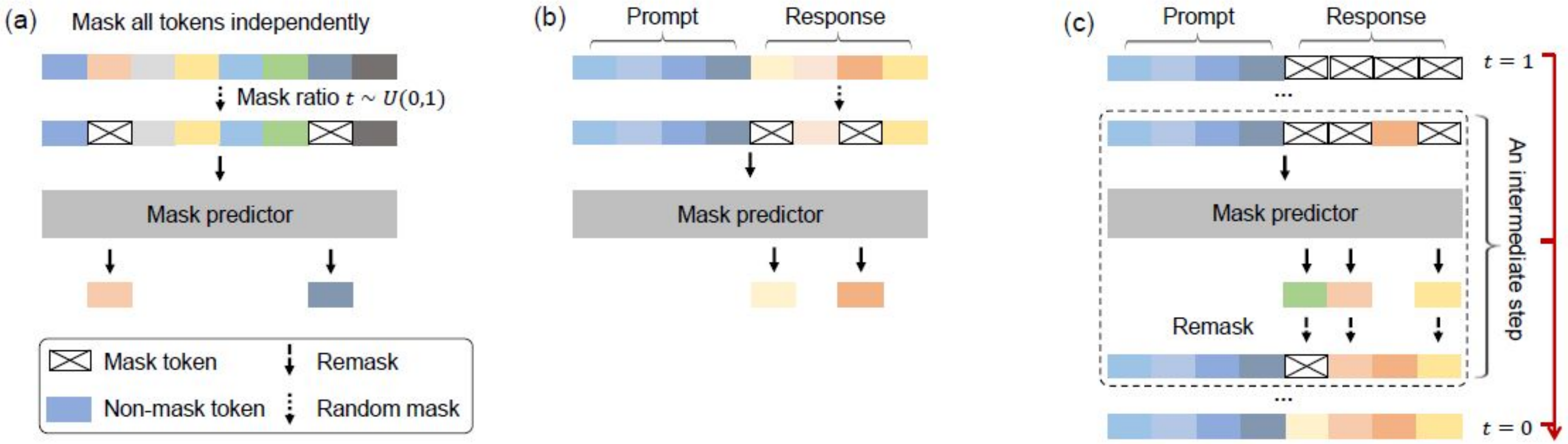
SINQ Llada Performance on General Tasks

Model Setting	Method	WinoGrande	PIQA	ARC-C	ARC-E	MMLU (5-shot)	Avg	Drop
LLaDA-8B FP16	–	69.9	74.6	46.4/45.9	71.1/72.6	65.7/65.8	65.5/65.8	-
W4A16g128	GPTQ	69.7	73.9	47.9	72.5	64.7	65.7	↑0.3%
W4A16g128	AWQ	67.3	70.3	44.5	73.4	65.6	64.2	↓2.0%
W4A16g128	SINQ	69.1	73.9	45.8	71.3	64.9	65.0	↓1.2%

SINQ Dream Performance on General Tasks

Model Setting	Method	WinoGrande	PIQA	ARC-C	ARC-E	Avg	Drop
Dream-7B FPModel-	–	68.4/69.0	74.4/74.9	59.0/59.7	83.1/84.3	71.2/72.0	-
W4A16g128	GPTQ	68.2	73.9	58.1	82.1	70.6	(↓0.8%)
W4A16g128	AWQ	65.2	69.6	55.8	82.0	68.2	(↓4.3%)
W4A16g128	SINQ	67.9	74.8	60.0	82.8	71.3	(↓1.0%)


(a)
Mask all tokens independently
Mask ratio $t \sim U(0,1)$
Mask predictor
Mask token
Remask
Non-mask token
Random mask
(b)
Prompt
Response
Mask predictor
(c)
Prompt
Response
$t = 1$
...
Mask predictor
Remask
An intermediate step
...
$t = 0$

Analyzing Quantization Error Per timestep

```
prompts = [
    "Lily can run 12 kilometers per hour for 4 hours. After that, she runs 6 kilometers per hour. How many kilometers can she run in 8 hours?",
    "Joy can read 8 pages of a book in 20 minutes. How many hours will it take her to read 120 pages?",
    "Randy has 60 mango trees on his farm. He also has 5 less than half as many coconut trees as mango trees. How many trees does Randy have in all on his farm?",
]
```

Prompt 2: Joy can read 8 pages of a book in 20 minutes. How many hours will it take her to read 120 pages?

<u>FP model Output</u>

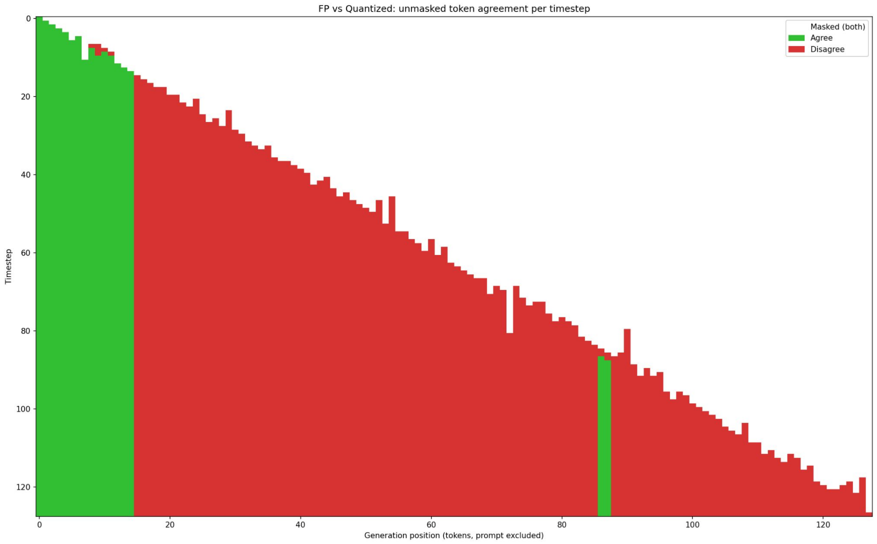
Joy can read 8 pages in 20 minutes, which is 8/20 = 0.4 pages per minute. To read 120 pages, it will take her 120/0.4 = 300 minutes.
Since there are 60 minutes in an hour, it will take her 300/60 = 5 hours to read 120 pages. The answer is 5

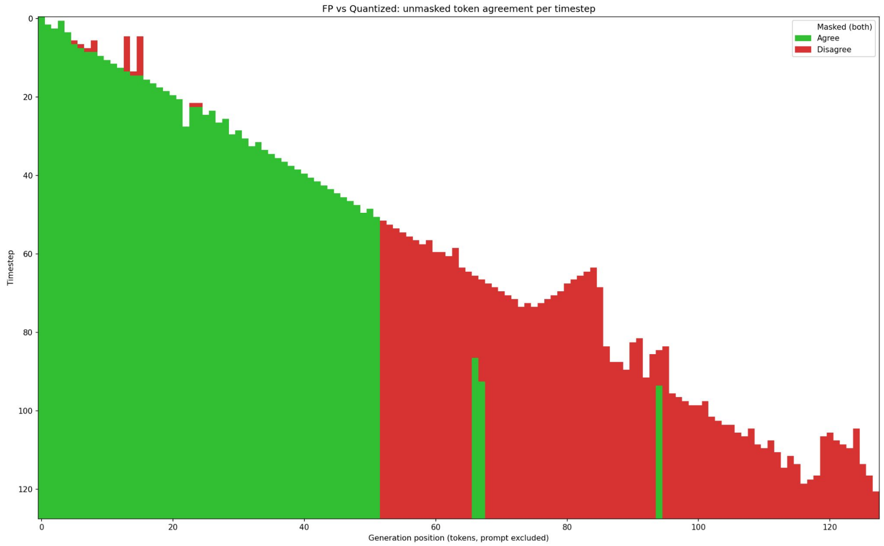
<u>Quantized model Output</u>

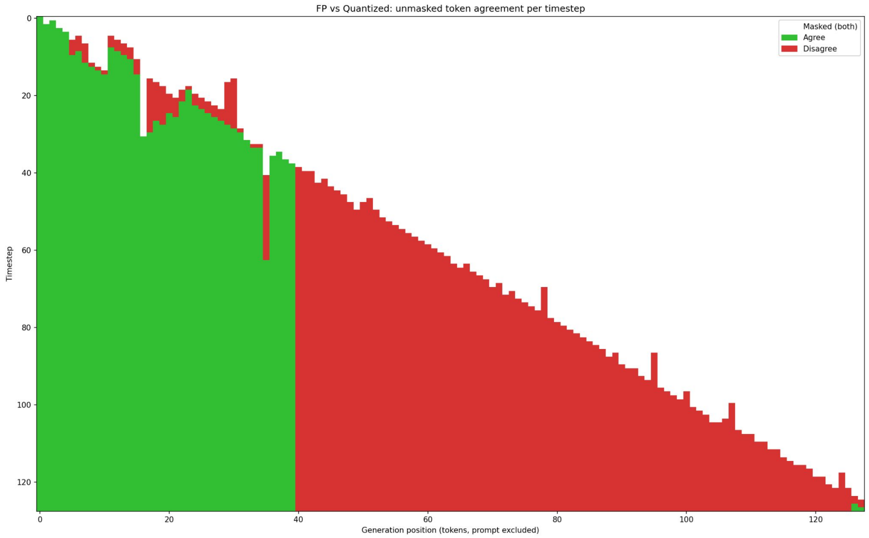
Joy can read 8 pages in 20 minutes, which is equivalent to 8 pages per hour (since there are 60 minutes in an hour).
To find out how many hours it will take her to read 120 pages, we divide the total number of pages by the number of pages she can read in an hour:
120 pages / 8 pages per hour = 15 hours. So, it will take Joy 15 hours to read 120 pages

Analyzing Quantization Error Per timestep

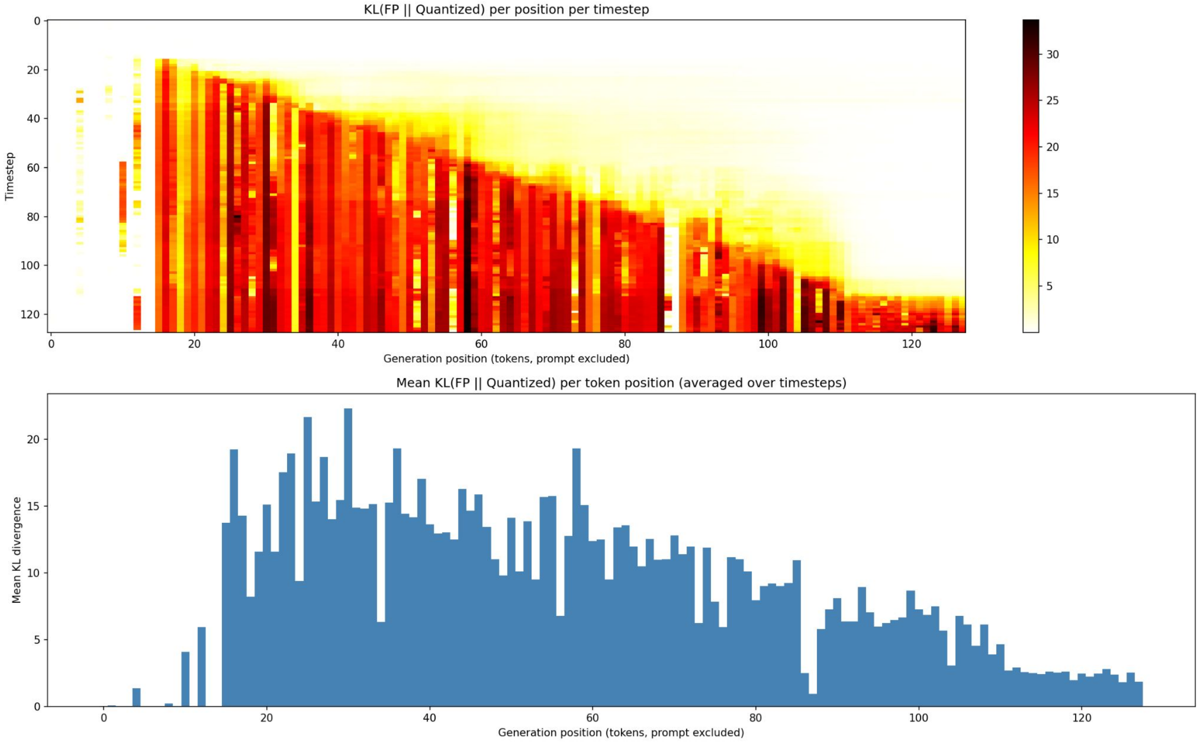
Llada unmasks one token at a time. This shows the disagreements on unmasked tokens at each timestep

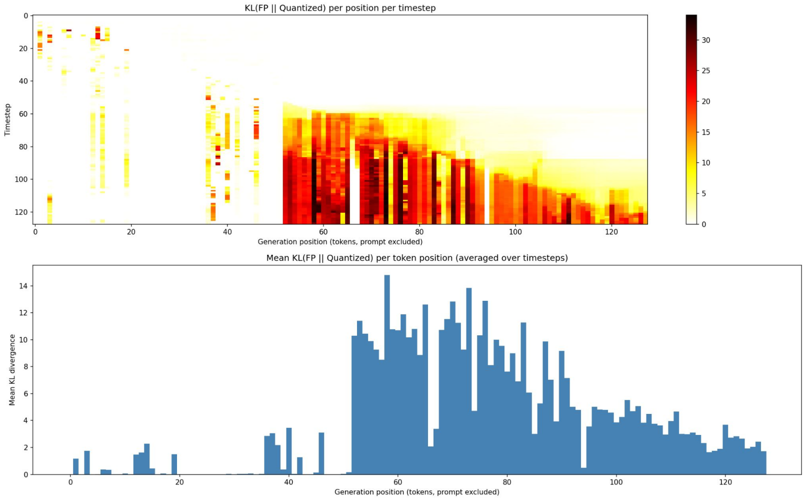


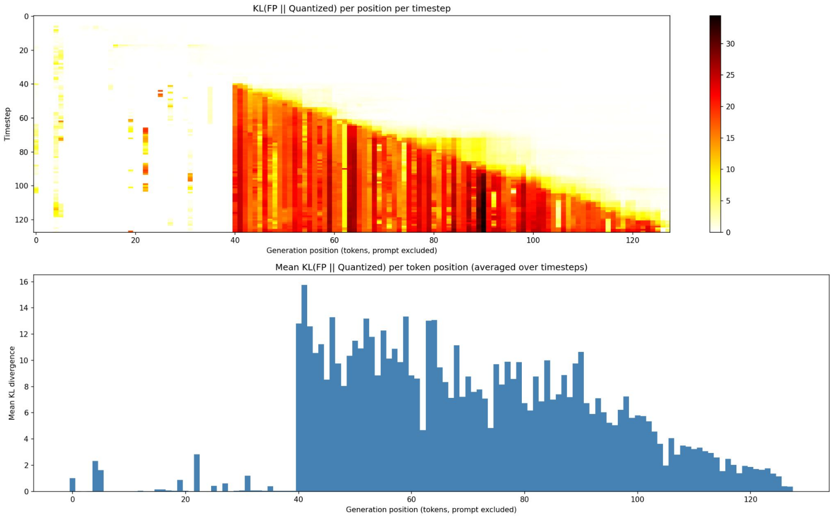

FP vs Quantized: unmasked token agreement per timestep
Masked (both)
Agree
Disagree
Timestep
Generation position (tokens, prompt excluded)


FP vs Quantized: unmasked token agreement per timestep
Masked (both)
Agree
Disagree
Timestep
Generation position (tokens, prompt excluded)

KL Divergence between FP & SINQ Quatnized Plots




KL(FP || Quantized) per position per timestep
Timestep
Generation position (tokens, prompt excluded)
Mean KL(FP || Quantized) per token position (averaged over timesteps)
Mean KL divergence
Generation position (tokens, prompt excluded)


KL(FP || Quantized) per position per timestep
Timestep
Generation position (tokens, prompt excluded)
Mean KL(FP || Quantized) per token position (averaged over timesteps)
Mean KL divergence
Generation position (tokens, prompt excluded)

Analyzing Quantization Error on General Questions

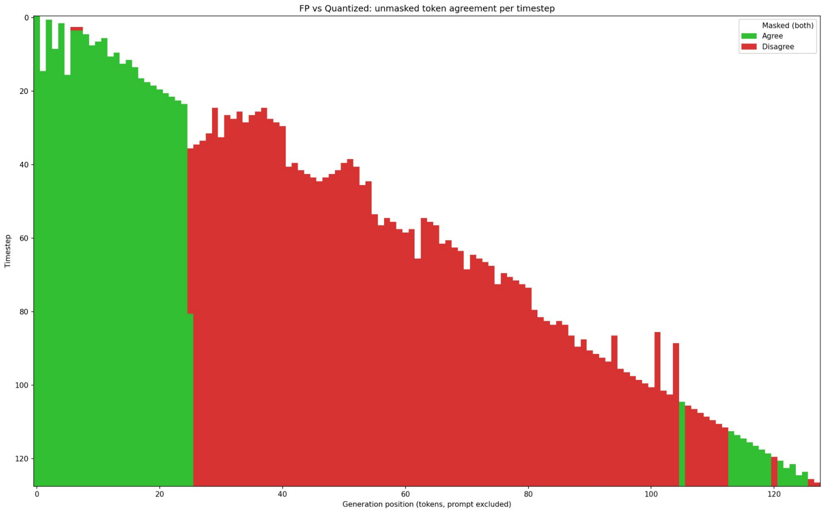
```
prompts = [
    "Write a short poem about the ocean at night, focusing on the sounds and feelings of standing at the shore.",
    "Explain how a neural network learns from data. Describe the forward pass, loss computation, and backpropagation in simple terms.",
    "Write a brief story opening: a detective arrives at a crime scene in a rainy city and notices three unusual clues.",
]
```

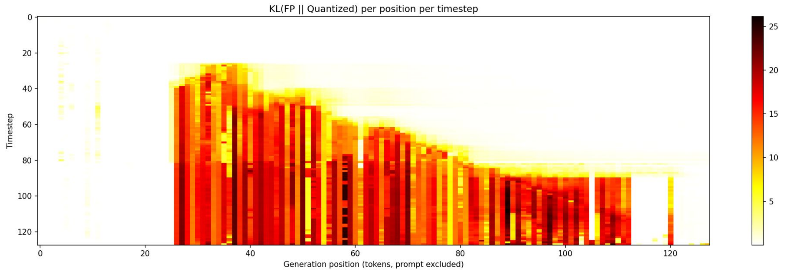
Prompt 3: Write a brief story opening: a detective arrives at a crime scene in a rainy city and notices three unusual clues.

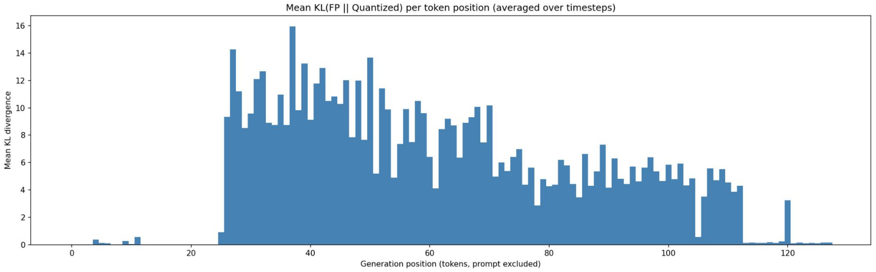
FP model Output A detective arrives at a crime scene in a rainy city and notices three unusual clues. The first clue is a small, rusty key lying on the ground. The second clue is a large, broken umbrella, lying open in the rain. The third clue is a small, torn flag, fluttering in the wind. The detective is puzzled by the significance of these clues and decides to investigate further.

Quantized model Output A detective arrives at a crime scene in a rainy city and notices three unusual clues. The first clue is a small, rusty metal box hidden in the corner of the room. The second clue is a series of strange symbols etched into the wall, and the third clue is a trail of blood leading to a hidden compartment in the wall. The detective is intrigued by the clues and decides to investigate further.

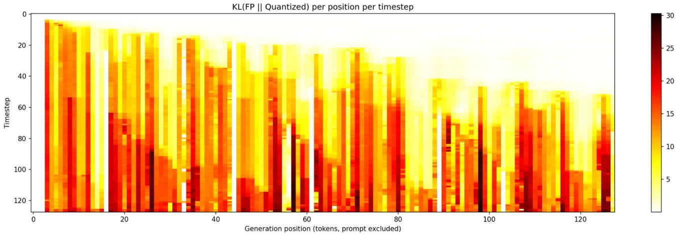
Plots for Prompt 3

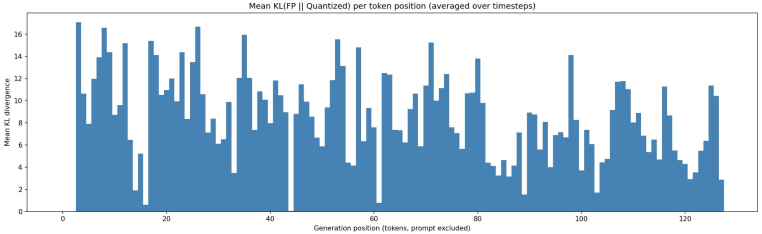


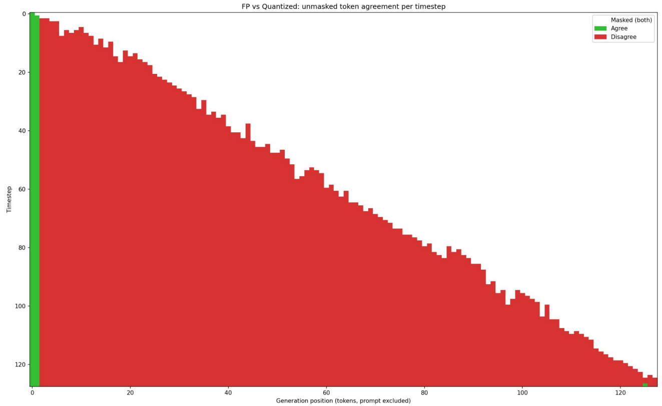


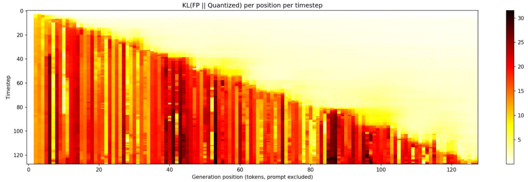


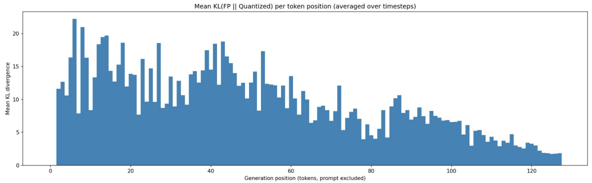
Plots for other Prompts











Key Takeaways & Next week Plan

- Quantization error is caused by having different confidence values
- Prompt type affects the robustness of quantization

Plan for next week:

- Report results for coding and math benchmarks
- Analyze error accumulation inside the model across layers
- Try to come up with a method to reduce the KL divergence across timesteps